



Course Project of Small Scale Polygeneration

Poly generation System Design and Optimization

**FRANCESCO QUERCIOLI
MICHELANGELO INZERILLI
PAPONSON INDRACHAI-EA
FELIX NAJITE OKENE
JIAOJIAO SONG
KIAN GROSSARD-AMIN**

Abstract

This project analyzed the process of designing and gradually improving a poly generation system to supply power for a sport center located in the city center of Brisbane, Australia. The total area of the sports center is 2,600 m², the facility includes one indoor and one outdoor Olympic-standard pools, both with a constant temperature heating need. Over half of the energy demand comes from the heating, the rest come from the electricity and ventilation cooling, the cooling is powered by an electrical cooler. The total annual demand is 500 kWh electricity and 698 kWh heating. Yearly round demand including heating, cooling and electricity is considered. Several scenarios are analyzed, including with and without energy battery storage systems, off and on grid.

The system design starts with an 88.2 kW PVs and 90 kW biomass CHP and finally optimized to a more comprehensive ploy system with a 120 kW PV system, a 90kWh CHP, an 80 kW battery and bidirection connected to grid. The energy balance analysis shows a strong seasonal variability and the important balancing role of the battery and grid connection.

The LCOE is decreased from 0.085, referred to a baseline case, to 0.070 €/kWh in the optimization process. When comparing with the case before implementing a poly generation system, our project allows 2 years payback. The final CAPEX is 336,416 euros with an OPEX 9,701 euros A business model and a sustainable assessment are also developed to evaluate the project's feasibility and social impact. The annual CO₂ emission is estimated at 3.535t.

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1.1 Objective of the Project

This project aims to address the objectives of SDG7[1] ensuring access to affordable, reliable, sustainable, and modern energy for all. It involves the design of a small-scale poly generation system in Brisbane, Queensland, Australia, to supply for energy for a community sports center and a swimming pool. Yearly round demand including heating, cooling and electricity is considered. Several scenarios are analyzed, including with and without energy battery storage systems, off and on grid. The primary aim of this project is to design and optimize the energy production system to achieve the most realistic and economically feasible while satisfying the hourly demand profile throughout the year. A business model and a sustainable assessment are also developed to evaluate the project's feasibility and social impact.

1.2 Project Scope and System Boundaries

This project focuses on simulating the hourly energy balance of a polygeneration system for the entire year of 2023 under various scenarios, such as with and without battery storage, and grid-connect and off-grid configurations. The systems boundaries are from receiving energy sources such as solar radiation, biomass and electricity, analysis the energy flow within system components, and finally generating required heating, cooling and electricity.

1.3 Methodology

A step-by-step working methodology is employed to gradually increase the complexity of the poly generation system. At each step, the hourly demand must be met, while total investment cost, Levelized Cost of Energy (LCOE), and payback period are used as indicators to evaluate the economic feasibility.

In the first step, the initial system is developed without energy storage. A CHP plant is designed to meet the demand when solar resources are unavailable. Surplus energy is kept as low as possible but remains partially unavoidable. In the second step, a battery is added to balance the supply and demand. The capacity of PV and battery are adjusted accordingly to achieve optimal economic performance. In Step 3, a bidirectional grid connection is introduced and the economic performance is recalculated to compare against off-grid scenario.

Further optimization proposals are explored, including integration of a small wind turbine system and adjustments to the generation capacity of key components. Finally, a comparative analysis is conducted to identify the optimal system design. A business model of implementation and sustainability assessment are also carried out to evaluate the system's technical, economic, environmental feasibility and social impact.

2 LOCATION SELECTION AND DEMAND DESCRIPTION

Brisbane receives approximately 2,792 hours of sunshine annually[2], providing ideal conditions for sporting activities and the operation of swimming pool facilities. The sporting facility was sized as a medium-scale complex with a total floor area of approximately 2,600 m², which corresponds to the standard size range for medium community sports centers. The facility includes two Olympic-standard pools, as shown in Fig. 1: a 50*25 m indoor pool with a uniform depth of 2 m, heated to 27.6 °C and equipped with a moveable bulkhead that allows it to be converted into two 25 m pools with advanced timing and scoring systems; and a 50*25 m outdoor pool offering 13 lanes with depths ranging from 4 m to 2 m, also heated to 27.6 °C, featuring diving towers and boards, and additionally used for Aerial Ski Jump training. The centre operates an average of 14 hours per day.

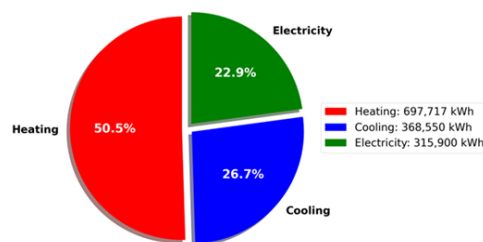


Figur 1. Brisbane sport center facilities

2.1 Electricity, Heating and Cooling Demand

Aquatic centers depend on both thermal and electrical energy. Thermal energy is used for space heating, pool water heating, and domestic hot water, while electrical energy powers cooling, ventilation, lighting, and mechanical equipment. On average, ventilation (including heating and cooling) accounts for about 45% of total energy use, pool water heating 33%, other space heating and ventilation 10%, lighting and power 9%, and domestic hot water 3% [3].

Studies in Australia show that aquatic centers are far more energy intensive than typical commercial buildings—often up to seven times higher—due to long operating hours and the constant need for water heating, ventilation, and pool circulation systems. A review of 22 aquatic centers in Victoria found annual energy use ranging from 648 to 2,283 kWh/m², with some facilities reaching 3,544 kWh/m²[4].



Figur 2. Energy consumption in sector

The electrical energy component was adjusted to better represent the operating conditions of a typical community aquatic facility in Brisbane. To account for the higher energy intensity of

such facilities, the electricity demand for lighting, pumps, fans, and plug loads was increased by a factor of three. This modelling assumption, informed by Australian benchmarks and climatic conditions, results in a total annual energy demand of approximately 1,198 MWh/yr, as summarized in Figure3

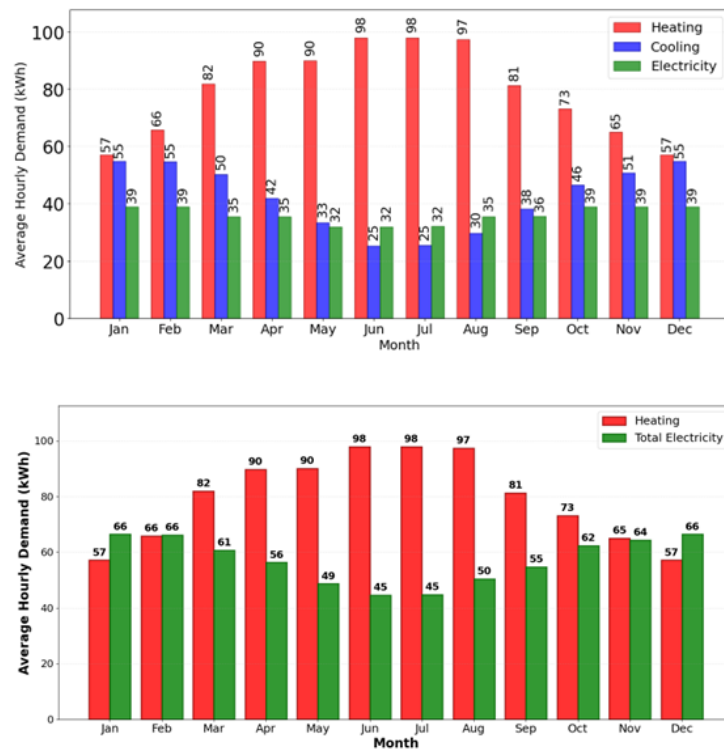


Figure 3. Monthly demand (upper: heating, cooling and electricity; down: merged electricity assuming COP=2)

The figures above show how energy demand changes throughout the year in response to the shifting seasons. Heating needs increases during the colder months, while cooling becomes more important when temperatures rise. Electricity use stays steady, with only small changes from month to month. When everything is combined, the total energy demand clearly follows the rhythm of the seasons, shaped mostly by how warm or cold the weather is.

2.2 Current Energy Situation

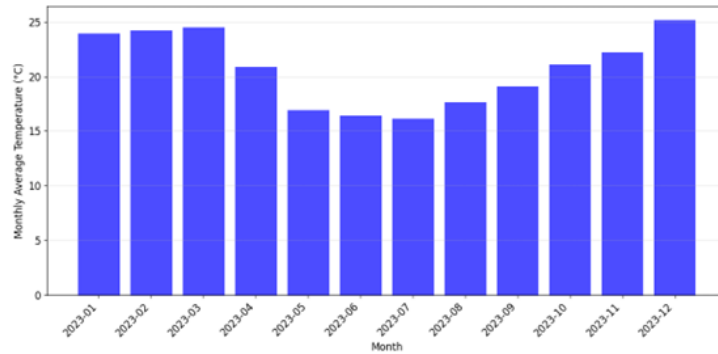
Brisbane receives electricity through the National Electricity Market via the Queensland high voltage network, planned by AEMO and operated by Powerlink. The 275 kV transmission backbone can deliver about 1,400 MW from Central Queensland to Brisbane during typical peak periods. Only Brisbane City Council operations are certified carbon neutral, achieved through energy efficiency measures, on site renewable generation, renewable energy purchases, and carbon offsets. Most electricity in Brisbane still comes from coal, which supplies about 64 percent of Queensland generation. Solar provides around 22 percent, gas about 7 percent, and wind and hydro only a small share. The remaining 1 percent comes from emerging sources such as batteries and biomass. Industrial electricity use is the largest source of emissions in Brisbane at roughly 5.37 Mt CO₂ e. Commercial buildings also contribute significantly, while residential emissions are lower due to widespread uptake of rooftop solar and reduced reliance on grid electricity.

3 ENERGY SUPPLY DESCRIPTION

Brisbane is located on Australia's eastern coastline. Its proximity to the Coral Sea and position within the subtropical zone shapes a warm and stable climate that is highly well-suited for renewable energy generation [5]. The region experiences a humid subtropical climate characterized by hot, humid summers and mild, dry winters, resulting in consistent solar exposure, moderate temperatures, and predictable seasonal weather patterns.

3.1 Temperature Profile

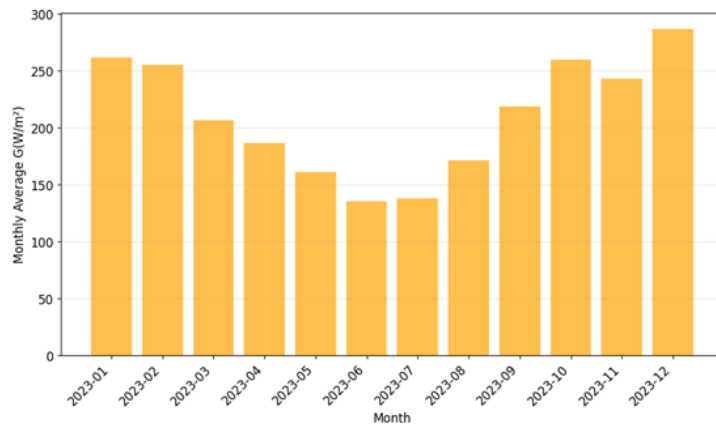
The temperature relatively warm conditions throughout the year, with average temperatures ranging from 16–29 °C across the seasons. Summer months (December–February) are typically warm and humid, accompanied by high solar radiation intensity. Autumn and spring provide mild transitional conditions, while winter (June–August) remains relatively dry with stable temperatures and clear skies. Rainfall is concentrated in the summer season.



Figur 4. temperature

3.2 Solar Energy

The global horizontal irradiance (GHI) exhibits strong seasonal variation. During the summer months (December–February), peak irradiance reaches between 1,000–1,100 W/m², supported by longer daylight hours and a high solar elevation angle result in high energy yield. In contrast, winter months (June–August) show lower maximum irradiance levels, typically around 600-800 W/m², with shorter daylight hours and lower sun position. This seasonal pattern demonstrates consistently high solar resource availability which is reflected in the annual solar irradiance profile in Figure 5.



Figur 5. Annual Solar Irradiance in the selected location in Brisbane, Queensland

3.3 Wind Resource

Average wind speeds typically range from 1–6 m/s at 10 m height with high variation throughout the year. These speeds are not suitable for utility-scale wind farms, but for distributed wind systems or community-scale installations[5].

3.4 Biomass Supply

Queensland region including Brisbane has a diverse and abundant biomass resource base driven by strong agricultural, livestock, and food-processing industries. The high-resolution GIS assessments indicate that biomass availability in the state is both geographically concentrated and seasonally structured [6]. Sugarcane is the dominant biomass, accounting for over 75% of the state’s total biomass potential, especially in coastal regions. Bagasse alone provides nearly 46% of all available biomass, with a strong energy yield and relatively stable supply [7]. Bagasse availability in Queensland is closely linked to the operational cycle of the state’s sugar industry, which is one of the largest in Australia. The annual sugarcane crushing season typically runs from June to December, during which sugar mills produce bagasse as a by-product of cane processing. This six-month period accounts for the majority of the resource supply, with peak availability occurring between September and November when mill throughput is highest. Outside the crushing season, January to May, bagasse production is minimal [8]. Therefore, storage solutions are required to maintain a continuous supply and ensure sufficient feedstock availability during this low-production period.

3.5 Supported Policies

The Queensland Government has established ambitious long-term targets through the Queensland Energy, which aims for 70% renewable energy by 2032 and 80% by 2035, supported by investments in grid upgrades, storage infrastructure, and clean-energy innovation. The state also implements mechanisms that incentivize renewable fuel use and indirectly encourages biore-source development [9]. In Brisbane and the broader South East Queensland (SEQ) region, rooftop solar consumers can participate in a market-based feed-in tariff (FiT) scheme. The FiT rates are offered directly by electricity retailers such as Origin Energy, AGL, and EnergyAustralia, each providing different export prices depending on the retail plan [10].

4 INITIAL SYSTEM DESIGN

4.1 Initial System Design

The initial step in the system design involved the conversion of the thermal cooling demand into an equivalent electricity demand using the minimum full-load COP for an air-cooled water chiller [11]. The result of the cooling electricity demand was then combined with the initial electricity demand to create the total electrical load for the next design phases.

Considering the favorable solar irradiance in Brisbane (average global horizontal irradiation: 153.94 kWh/m²)[12], a roof-mounted photovoltaic (PV) system with optimal tilt angle and azimuth was selected as the first source of energy generation. The nominal power installed was determined by searching for the highest hourly electrical demand in the dataset and matching it with the installed peak capacity, which accounts for 88.74 kWp. The number of panels installed for the Trina Solar TSM340 PE14H model is then 261, each rated at 340 Wp. An hourly analysis of production was later performed using the online PVGIS tool, which allowed the definition of total system losses at 13.90% based on the following components: 3% shading losses, 10% temperature losses, 4% other losses, 17.1% of module efficiency [13], 97% inverter efficiency (according to the manufacturer data, the maximum efficiency is 98.8%)[14]. For inverter selection, a single 75 kW unit was assumed at the system output, since the maximum energy produced by the photovoltaics is 72.8 kWh. For this matter, the SMA Sunny Highpower Peak1 model was chosen[14].

A PV system alone is not sufficient to meet the electrical demand or contribute to the heating production. Therefore, a biomass-based combined heat and power (CHP) system was designed to satisfy the requirements. The design process proceeded similarly to the PV's, by initially finding the peak electricity and peak thermal demand, equal to 88.4 kW(e) and 146 kW(th) respectively. Dividing the two values gave a reasonable heat-to-power ratio of 1.65, falling within the typical range for CHP plants[15]. To ensure that the whole system's demand is met, the CHP system was assumed to operate continuously, with a rated power of 90 kW(e) and 150 kW(th).

The bagasse is considered as the biomass fuel for CHP system. It is utilized as fibrous by-product from sugarcane crushing in the sugar industry. Owing to its high availability and relatively low cost in Queensland, bagasse with an average moisture content of 45% and a lower heating value (LHV) of 7,413 kJ/kg is considered as a primary energy source [16]. This moisture level represents the typical condition of bagasse directly obtained from sugar mills, allowing stable combustion performance without requiring additional drying process. The required amount is calculated from the annual heating demand, the 85% thermal efficiency of CHP, which result in approximately 744 tons per year of biomass [17]. At an average market price of 40 AUD/ton in 2020, the corresponding annual biomass cost is estimated at 29,743 AUD per year [18].

Ultimately, the initial design is represented in Figure 6 below. There is a 88.2kWp PV system (426m²) and a 120kW(e) biomass CHP (which will only serve 90kW(e)), with a heat to power ratio of 1.65.

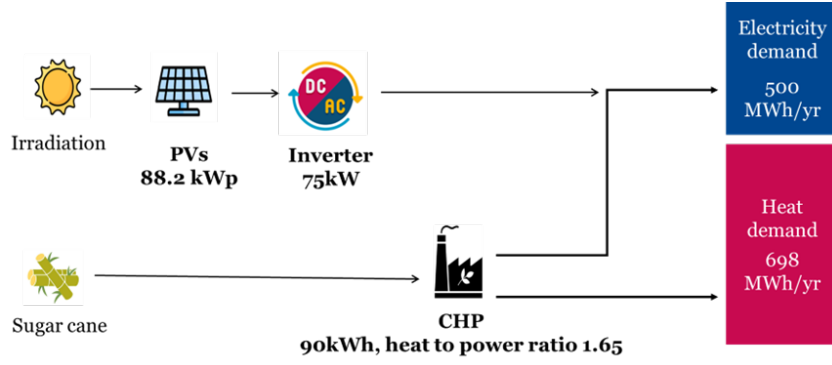


Figure 6. Initial system design sketch

4.2 Energy Balance Analysis

Looking at the share of the energy produced from each source, the CHP was found to produce around 74% of the total yearly electricity, while the PVs were found to produce around 26%. The CHP supplies the total heat demand. It is also interesting to note that the electricity produced by PVs is sometimes higher than the demand, which means some energy gets curtailed. This will however be discarded for now and mentioned in the following sections.

The energy balance can be visualized in the following figures. A typical summer and winter week have been selected to observe the differences. What can be observed is the higher electricity production by the PVs during the summer, supplying the total demand for several hours. It is also interesting to note that the overall demand is higher during summer than winter.

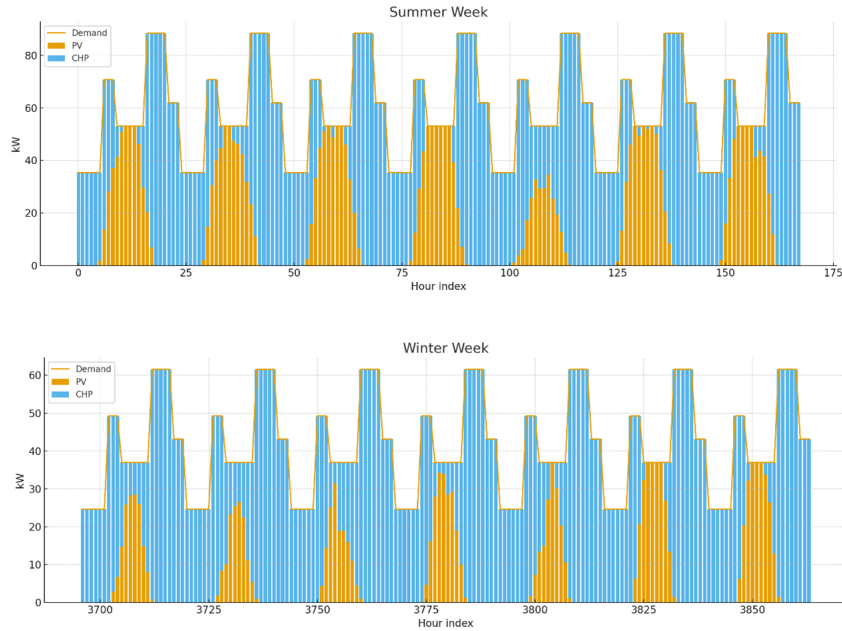


Figure 7. Power source stack figure

4.3 Economic Analysis of this Initial System

Summarizing again the energy system configuration, it comprehends 261 solar PV panels (Trina Solar TSM340 PE14H) coupled with a 75 kW inverter (SMA Sunny Highpower Peak1), a CHP (ecoGEN-120SB) with a rated electrical plus thermal output of 339 kW and an Air-Cooled

chiller (Conquest CGAX Air-Cooled Chiller). The reference currency is euro (€); if conversion was needed the following factors were used: 1\$ equals to 0.86€ and 1AUD to 0.56€. Particularly, each PV unit has a cost of 144€ [19], which consists of a total cost of 37,584€ when considering 261 panels. The system comprehends the above cited inverter, with a unit cost of 4,719€ [20]. The total CAPEX will account for 42,303€, while the OPEX is approximated to the 3% of the CAPEX and results to be 1,269€ per year. Furthermore, the CHP, which costs can be calculated considering a CAPEX equivalent to 1,863 €/kWe and an OPEX of 0.015 €/kWh [21], result to be 223,583€ and 12,000€ respectively. The OPEX excludes fuel costs, which account for roughly 16656€ per year. The last piece of equipment to consider is the chiller. The price is of roughly 29,497€, while the OPEX 885€. CAPEX was calculated following 1,500 \$/tons[22] and OPEX as 45 \$/tons[23] The following graph (Figure 8) represents how each equipment is influencing the total system's costs.

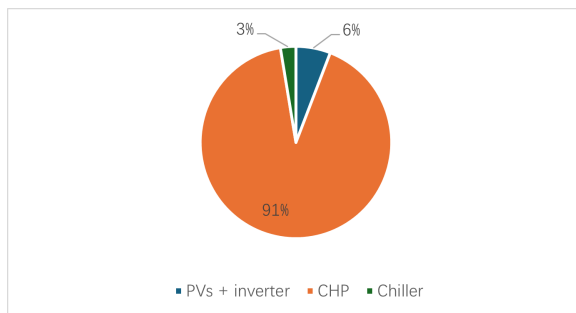


Figure 8. Total costs contribution

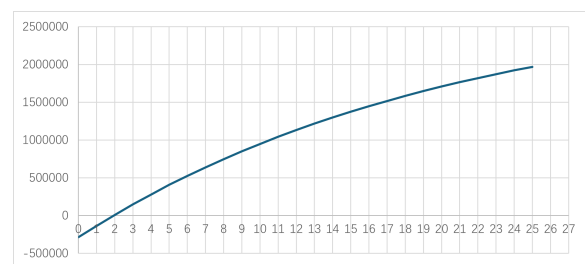


Figure 9. NPV compared to baseline

As can be seen, the major contributor to the high costs of the system is the CHP, accounting for a 76% influence in CAPEX and 88% in OPEX and fuel costs. Finally, considering an interest rate of 5%, and an optimistic lifetime of the equipment of 25 years, the following KPIs were calculated: relative Net Present Value (NPV), Return of Investment (ROI) and Levelized Cost of Energy (LCOE), which equation can be found in Appendix 1. To evaluate an NPV comparison, a baseline scenario has been established. It comprehends the purchasing of all the electricity demand from the grid, for a total of 136,756€ and the supply of heat from a natural gas boiler, which has a capacity of 150 kW, a CAPEX of 7,409€ and a yearly fuel cost of 58,204€ considering a stable price of gas of 0.08 €/kWh.

Figure 9 represent the cumulative NPV difference trend to the baseline, during the 25 years, where y-axis values are in euros. As can be seen, even with an off-grid and not optimized case, it is still reached a ROI in the second year compared to the baseline scenario, which has extremely high fuel (natural gas) and electricity costs and only serves as a worst-case scenario comparison.

Finally, the LCOE resulted to be of 0.07 €/kWh, with an annual energy production of 1,382,167 kWh. Further analysis will be made and the different KPI's will be confronted to evaluate the best system configuration for the given location.

5 INTEGRATION OF BATTERY ENERGY STORAGE

5.1 System Includes a Battery

As next step of the system design, an 80 kWh battery storage system was added, so that the self-consumption of the PVs would increase. The battery chosen is the Lithium battery Freedom Lite 80 kWh [24]. The inclusion of a BESS of this size allows the system to receive emission-free electricity for 39% of the time, while before, without storage, it was only 26%.

The remaining 293.4 MWh of electricity demand, along with the entire thermal demand, is still met with the use of the CHP system, which remained untouched. This choice guarantees that the demand is always met, at the cost of generating more electrical surplus. The flexibility of the CHP system will be addressed later in the final optimization.

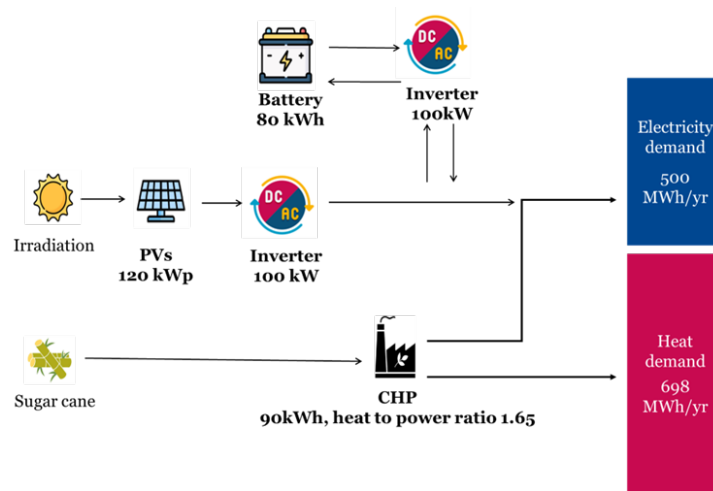


Figure 10. System includes a battery

5.2 System Size Adjustment

To further enhance PV self-consumption, the rated capacity increases to roughly 98 kW (290 panels) in this scenario. This change will increase the inverter size as well to a 100 kW ZCS AZZURRO 3PH 100KTL-V4 unit, sized for the maximum energy output at 98.44 kWh. Its maximum efficiency of the inverter is 98.6% [25], which is still above the value used previously (97%), so no adjustment in the efficiency was needed. This power output growth increases the unused energy but improves the self-consumption of the whole system.

5.3 Energy Balance Analysis

The implementation of the battery significantly increases the share of electricity generated by PVs in total electricity generation. Also considering the battery increases the share of PV to 35%, leaving the CHP responsible for generating 65% of the total yearly electricity demand. While it could be suspected that the electricity provided by the PV might not reach demand during the winter, that is not the case. In fact, thanks to the oversizing of the PV system, the battery surprisingly stores excess PV electricity during the winter, as can be seen in the following figure. This can be explained by the fact that the electricity demand falls during winter.

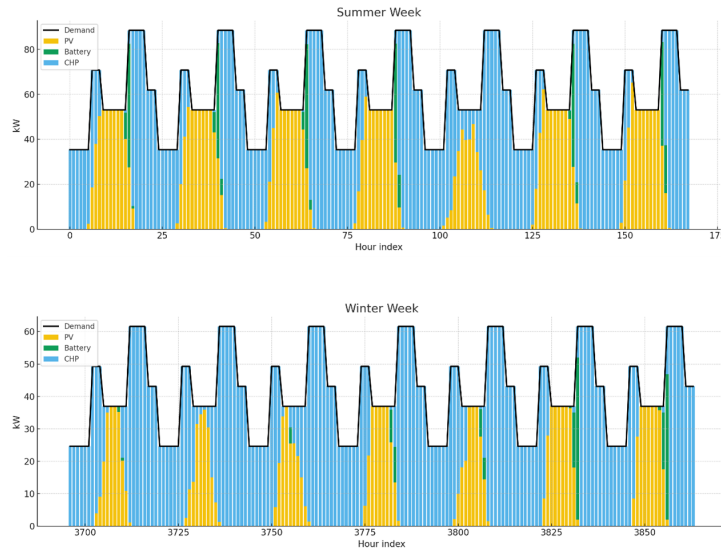


Figure 11. Power source stack figure (battery included)

5.4 Local incentives or support mechanisms

The Australian Federal Government’s “Cheaper Home Batteries Program”[26], launched on 1 July 2025, provides an upfront rebate for new battery installations paired with solar PV systems. Australian households, businesses and community organizations can get a discount on the upfront cost of installing small-scale battery systems. The discount is generally provided through accredited installers/retailers. The discount will be offered as an upfront reduction on the cost of the battery. Batteries from 5 kWh to 100 kWh nominal capacity are eligible under the Program. Small-scale Technology Certificates (STCs) will only be provided for the first 50 kWh of the usable capacity of a battery system. Batteries with a capacity larger than 50 kWh and up to 100 kWh can still receive STCs, up to and including 50 kWh of its usable capacity, but not beyond. The discount rate in 2025 is 9.3 STCs per kWh of usable capacity, equivalent to around \$372 per kWh. The discount will represent around 30 per cent of the cost of an installed battery in 2025 (for example, around \$3,000 discount on the cost of installing an 8 kWh battery at \$10,000). The number of STCs will decline for battery systems until the scheme ends in 2030. The calculation of STCs will be based on the number of STCs per kWh of installed usable capacity of the battery and the year of installation, as outlined in Table 1.

Tabell 1. Battery incentives in Australia

Year	CAPEX (Australian dollars)	STC factor
2025	372	9.3
2026	336	8.4
2027	296	7.4
2028	260	6.5
2029	224	5.6
2030	188	4.7

5.5 Economic Analysis of this Battery-included System

The most relevant changes in this scenario are the implementation of a battery system of 80 kWh capacity and the change of the previous 75kW inverter to a 100 kW one, together with the new

total of 290 PV units. Firstly, focusing on the battery, its investment cost was approximated as 5,000 SEK/kWh (450 €/kWh) for a total of 36,000€, while the OPEX as its 2.5% and so roughly 900€. On the other hand, the new inverter has a cost of 5,576€, compared to the 4719€ of the previous configuration.

The new investment and operational costs are reassumed in the table below.

Tabell 2. CAPEX and OPEX of system components

Component	CAPEX (€)	OPEX (€)
PVs + inverter	47,336	1,420
CHP	223,583	16,000
Chiller	29,497	885
Battery	36,000	900

Looking at the NPV comparison to the Step 1 system, it now also considers the installation and OPEX costs of the battery and of the increased number of PV modules, and their not optimized behavior. Saving in the OPEX are related to the minor use of the CHP, which led to less fuel and OPEX costs. As can be seen, a ROI in less than four years is achieved.

The LCOE of the system is of 0.067 €/kWh for a total energy production of 1,300,207 kWh.

6 INTEGRATION OF GRID

6.1 System Connected to Grid

In this step, the system is connected to the grid to achieve more dynamic and economic energy control. The electrical surplus is sold to the grid, and during peak load time, the shortage of electricity is imported from the grid.

6.2 Electricity Price

The hourly electricity price is referenced from the regulated retail electricity prices in regional Queensland 2023–24 report published by Queensland Competition Authority[27]. As indicated in Tab. The community sports center is classified under small business Tariff 22C, Band 5, due to annual electricity consumption exceeding 80 MWh, and high consumption during the day with PV energy export to the grid. Both buying and selling electricity prices are calculated dynamically using Time-of-Use structure regarding the operating hours in three different periods. The import electricity price (buying) is considered with relative tariff of 22C buying price as follows

- Off-peak (09:00 – 16:00 daily): 0.10875 AUD/kWh
- Peak (16:00 – 21:00 weekdays only): 0.63837 AUD/kWh
- Shoulder (non-peak, non-off-peak hours): 0.25632 AUD/kWh

The export electricity price (selling) is based on a Time-of-Use Feed-in Tariff (ToU-FiT) rate according to National Electricity Market (NEM) wholesale energy price[28] as follows

- Off-peak (09:00–16:00 daily): 0.03 AUD/kWh
- Peak (16:00–21:00 weekdays only): 0.1 AUD/kWh
- Shoulder (non-peak, non-off-peak hours): 0.05 AUD/kWh

6.3 Energy Balance Analysis

The share of PV and battery is the same as for the previous step, as the capacities of both technologies are the same, supplying 35% of the demand, leaving the CHP responsible for generating 65% of the total yearly electricity demand. The batteries' role is essentially to maximize the use of solar energy within the system.

An interesting thing that can be observed is the substantial amount of surplus present in the system. Indeed, it is completely self-sustainable and sells 53.66% of the total amount of produced electricity to the grid. This helps to recover the cost of the system.

The main difference that can be observed between the winter and summer is the fact that excess production usually occurs during the winter. During the summer, the spike in excess production is due to the production of solar electricity.

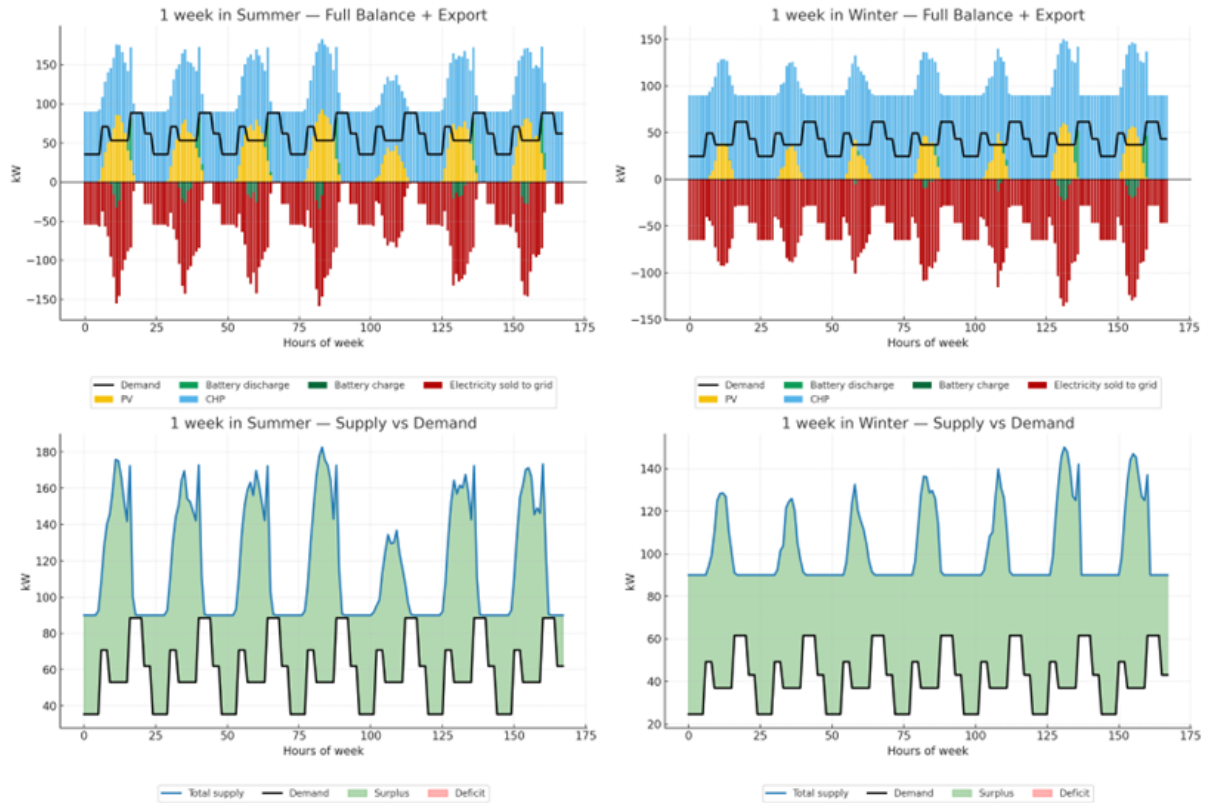


Figure 12. Energy balance of system with grid interaction

6.4 Economic Analysis of this Grid-connected System

This Step's economic analysis, as well as considering the new system's equipment sizes, also ponders the interaction with the grid.

Specifically, now the system is able to generate revenues from the grid, which account for a total of 11,212€ yearly, helping the economics of the system, which overall remain the same in equipment's size.

The revenues obtained from selling the on-site produced electricity make the system's NPV higher than the previous configuration.

7 SYSTEM FINAL OPTIMIZATION

The logic of the system's optimization is to minimize thermal surplus without additional CAPEX (i.e., avoiding auxiliary boilers). Two key changes were made: CHP operational flexibility and CHP-battery integration.

The CHP, left at the same capacity of 90 kWh(e) and 150 kWh(th), operates with a turn-down ratio of 30% [29], modulating its output down to 30% of the nominal capacity. This enabled the system to be more flexible and drastically reduce the thermal surplus, while providing sufficient heat at every hour. The second change, instead, is the implementation of the electricity coming from the CHP to the battery operations, and not just the PV output.

The system's operation, which is now heat-led, matches in the first place the thermal demand, while eliminating constant full-load operation. The control strategy now prioritizes:

1. CHP heat-led dispatch: CHP output depends on the hourly heat demand, producing proportional electricity (constant Heat-to-Power ratio at 1.65).
2. Surplus management: If the electric power coming from the CHP and the one coming from the PVs is higher than the demand, the excess goes to the battery. When this one is charged at its higher SOC limit, the remaining surplus is exported to the grid.
3. Deficit coverage: the battery discharges electricity if the production (PV+CHP) is not enough to meet the electrical demand, until the battery SOC reaches its lower limit (5 kWh). When this happens, the electricity is imported from the grid.

Ultimately, this solution enables the system to strategically use the battery by staying within the imposed limits of the SOC. The changes done were made to minimize thermal surplus, since the heat cannot be sold, without adding new CAPEX. On the other hand, the strategy increases the electricity surplus, which can later be sold to the grid and still be considered positive (revenue) in the cash flow.

7.1 Energy Balance of the Optimized System

An overview of the total energy (electrical) balance and the total electrical surplus can be observed in the following figure:

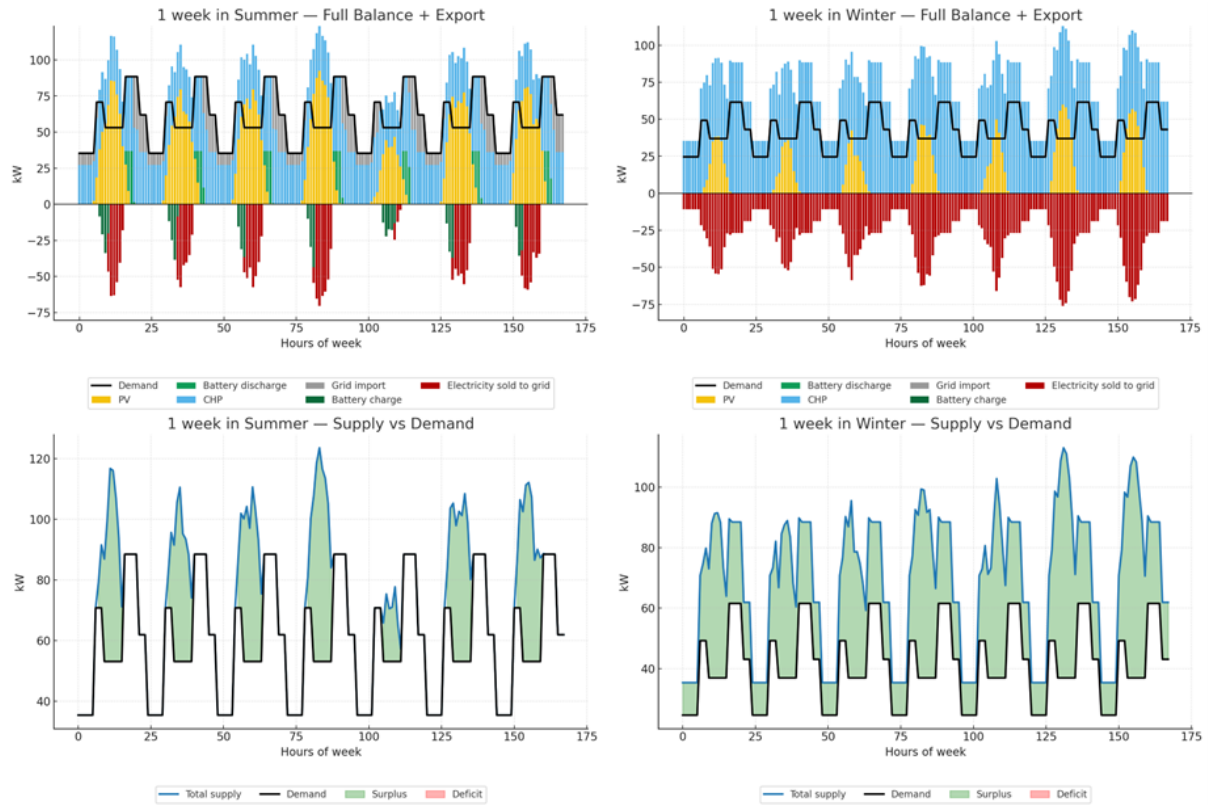


Figure 13. Energy balance of optimized solution

7.2 Economic Analysis of the Optimized System

The economics of the optimized system achieve the best results compared to the other steps when looking at the NPV. The costs reduction is caused by an optimized usage of the CHP which is linked to lower fuel costs. In fact, even if in this case the grid is used more as a source of electricity and the expenditures are higher than the revenues, the NPV is still higher than the previous step. The cumulative NPV difference trend is therefore shown in the following figure.

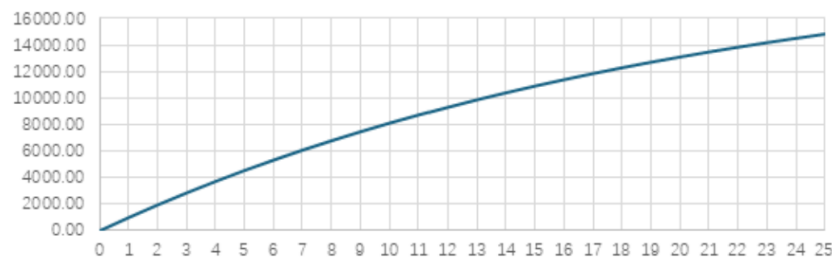


Figure 14. NPV difference before and after optimization for the grid-connected system

In this last optimized case, the LCOE accounts for 0.07 €/kWh. The final capex is 336,416 euros and final opex is 9701 euros.

8 CROSS COMPARISON OF ALL SYSTEMS

The baseline configuration relies entirely on grid electricity and a gas boiler to meet 452.40 MWh of electricity and 697.72 MWh of heat annually. It incorporates no local renewable generation and no surplus production by design. It is operationally straightforward but associated with high operating expenditures and carbon-intensive supply. Therefore, it serves as the reference case for all subsequent polygeneration designs.

In the **initial system**, an 88.74 kW PV system and a 90 kW biomass CHP unit are introduced. The CHP operates continuously, fully covering the heat demand and supplying 65% of the electricity demand, while PV provides the remaining share. This configuration already reduces the LCOE to approximately 0.07 /kWh and achieves a payback of about two years relative to the baseline. However, the lack of flexibility in CHP operation generates very large annual surpluses: around 603.80 MWh of heat and 461.24 MWh of electricity, so that nearly half of the total generated energy is not effectively utilized.

The **battery-included system** increases the PV capacity to 120 kW and adds an 80 kWh battery. The battery serves to increase PV self-consumption within the system. From an energy perspective, the thermal surplus remains unchanged at 603.80 MWh, while electrical surplus increases slightly to 486.91 MWh due to the larger PV system.

The **grid-connected system** maintains the same technical configuration but connects the system to the grid. Surplus electricity, still about 486.91 MWh per year, is now exported instead of curtailed, reducing overall costs. This system achieves an LCOE of around 0.05 /kWh. The thermal surplus, however, is unaffected by grid interaction and remains high.

The **final optimized system** introduces a 30% turndown ratio for the CHP and integrates its electrical output into the battery control strategy. CHP operation becomes heat-led, meaning its output is modulated to match the thermal load within the minimum load constraint. Any excess electricity is first directed to the battery and only then exported to the grid. This control strategy reduces annual thermal surplus from 603.80 MWh to 6.40 MWh and lowers electrical surplus to 166.32 MWh, while still satisfying the full annual heat and electricity demands. Biomass consumption and associated operating costs fall accordingly, and although grid imports rise, they are concentrated in low-tariff periods so that overall economic performance remains superior to the previous configurations.

In terms of surplus fraction (total surplus divided by total generation), the three intermediate polygeneration cases (**initial system**, **battery-included system**, and **grid-connected system**) all lie at roughly 48%, whereas the **final optimized system** reduces this value to about 13%. Taken together, the comparison shows that the combination of CHP flexibility, storage and grid interaction in the **final optimized system** simultaneously minimizes energy waste, maintains a low LCOE and yields robust economic performance, making it the most suitable configuration for the Brisbane community sports center.

9 BUSINESS MODELS OF IMPLEMENTATION

The Energy-as-a-Service (EaaS) business model by an Energy Service Company (ESCO) was selected as the most viable option. Although the polygeneration system offers significant energy savings and emission reductions, its complexity creates challenges for the sport center owner. For example, CHP requires specialized operation and continuous monitoring, PV and battery systems require dynamic optimization against real-time electricity pricing, and the demand supply needs to be managed for reliability operation.

The EaaS business model overcomes these barriers by transferring financial, technical, and operational responsibility to a specialized energy service provider (ESCO). It helps the owner to reach optimized energy generation, lowest operation cost, and less carbon emission without upfront investment need. The CAPEX and operational responsibilities will be transferred to ESCO. They will benefit from multiple revenue streams including tariff arbitrage and fuel contracting efficiencies. There are operational strategies for the business model to be aligned with the local context as follows

- **Dynamic Electricity Market Conditions**

Prioritize the PV energy self-consumption, schedule CHP operation during high retail electricity prices, and use the battery to store surplus energy and discharge during peak-tariff hours to minimize grid dependence and reduce operational costs.

- **Biomass Secure Supply**

Secure the multi-year contracts with bagasse supplier for the fixed and lower bulk price, manage the seasonal storage to ensure continuous supply, conduct the quality control of bagasse moisture content to ensure combustion efficiency and energy yield.

- **Forecast Demand Profile**

Predict the load consumption based on user behavior from membership analytic data and operating schedules to achieve the optimum demand-supply management.

- **Local Policy Alignment**

Ensure that the system design and long-term implementation plan align with municipal (Brisbane) through regional (Queensland) policies on carbon emissions, renewable energy integration, and decarbonization milestones.

10 SUSTAINABILITY ASSESSMENT

The proposed polygeneration system supports multiple Sustainable Development Goals: SDG 7 (clean energy), SDG 9 (innovation), SDG 11 (sustainable communities), and SDG 13 (climate action).

The environmental impact of the proposed polygeneration system presents significant benefits alongside manageable challenges. On-site solar PV generation reduces dependence on fossil-fuel-based grid electricity, decreasing operational greenhouse gas (GHG) emissions. Life cycle assessments confirm that solar generation has a much lower carbon footprint than fossil fuel alternatives. Enhanced ventilation and electrical cooling improve indoor air quality in the humid pool environment, reducing mold and chloramine formation risks.

GHG emissions are associated with electricity generation, grid consumption, and thermal energy production from the CHP using bagasse as biomass fuel. Australian regulatory datasets provide the reference emission factors for calculations.

- **Electricity CO₂ emission.** Accurate estimation requires verified region-specific emission factors. For Queensland, the official location-based electricity emission factor for 2024–25 is 0.71 kg CO₂-e/kWh, published by the Australian Clean Energy Regulator. This reflects Queensland’s actual carbon intensity rather than the broader National Electricity Market average, providing appropriate conservative calculations. PV generation is treated as zero operational emission.
- **Heating CO₂ emission.** According to Australian NGER (National Greenhouse and Energy Reporting) factors for bioenergy systems, residual non-biogenic emissions (CH₄ + N₂O) from bagasse combustion are calculated using:

$$EF_{\text{bagasse}} = 9.6 \text{ GJ/t} \times 1.4 \text{ kg CO}_2\text{-e/GJ} = 13.44 \text{ kg CO}_2\text{-e/t}$$

- **Total CO₂ emission.** With annual bagasse consumption of 263 tonnes, thermal energy emissions are approximately 3.535 t CO₂-e/year. Total operational carbon emissions, assuming no grid electricity imports, are determined solely by bagasse combustion at approximately 3.535 t CO₂-e/year.

Negative environmental aspects include embodied impacts from PV panel manufacturing and decommissioning, involving material extraction for components like glass and metals. Additionally, poor pool water management could worsen humidity and chloramine formation. Proper design and operation are crucial to maximize net environmental benefits.

The polygeneration system offers substantial social benefits with some manageable challenges. Visible renewable energy installations enhance community awareness and promote sustainability education, while improved thermal comfort, humidity control, and air quality benefit all users, including vulnerable populations. Community-level projects foster local engagement and strengthen social cohesion. However, potential concerns include noise from installation activities, temporary operational disruptions, and the risk of community resistance if stakeholder engagement is poorly managed. These can be mitigated through transparent communication, careful scheduling, and inclusive community involvement throughout the project lifecycle.

11 CONCLUSION

The research proves that a polygeneration system which combines PV with biomass CHP and battery storage and grid connection can fulfill the power and heat requirements of a Brisbane community sports center with swimming pool while achieving better energy savings and emission reductions than standard grid-and-gas-boiler systems. The development process begins with an off-grid system that produces excessive heat before reaching an optimized solution which uses CHP flexibility along with battery and grid integration to minimize energy waste and maximize renewable self-consumption while keeping LCOE competitive. The economic evaluation reveals that the optimized system provides fast payback durations and higher net present value while the sustainability assessment demonstrates its ability to support Sustainable Development Goals through increased clean power usage and decreased operational emissions and its ability to enable community participation through visible renewable systems and Energy-as-a-Service offerings which make facility operation costs more affordable.

12 REFERENCES

- [1] Eurostat, “SDG 7 - Affordable and clean energy,” 2025, Accessed: 2025-05-07. URL: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=SDG_7_-_Affordable_and_clean_energy.
- [2] “Sunshine & Daylight Hours in Brisbane, Queensland, Australia.” Accessed: 2025-11-12.
- [3] E. Trianti-Stournam. fl., “Energy conservation strategies for sports centers: Part B. Swimming pools,” *Energy and Buildings*, årg. 27, nr 2, s. 123–135, april 1998. DOI: 10.1016/S0378-7788(97)00041-8.
- [4] J. J. Duverge, P. Rajagopalan, R. Fuller och J. Woo, “Energy and water benchmarks for aquatic centres in Victoria, Australia,” *Energy and Buildings*, årg. 177, s. 246–256, okt. 2018. DOI: 10.1016/j.enbuild.2018.07.043.
- [5] European Commission, Joint Research Centre (JRC). “PVGIS – Typical Meteorological Year (TMY) Data Tool.” Photovoltaic Geographical Information System. Accessed: 2024-05-07, European Commission, Joint Research Centre. URL: https://re.jrc.ec.europa.eu/pvg_tools/en/#TMY.
- [6] Queensland Government. “Queensland Biomass Resource Assessment and Spatial Availability.” No publication date. Accessed: 2025-05-21.
- [7] Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES). “Australian Biomass Resources and Bioenergy Potential.” No publication date. Accessed: 2025-05-21.
- [8] Sugar Research Australia. “Australian Sugar Industry Production Cycle and Bagasse Availability.” Technical Report. No publication date. Accessed: 2025-05-21.
- [9] Queensland Government. “Queensland Energy and Jobs Plan Delivers Renewable Energy Targets and Clean Energy Investment.” Accessed: 2025-05-21. URL: <https://statements.qld.gov.au/statements/100137>.
- [10] Queensland Competition Authority. “Solar Feed-in Tariff for Regional Queensland 2023–24: Final Decision.” Accessed: 2025-05-21.
- [11] F. W. Yu, K. T. Chan, R. K. Y. Sit och J. Yang, “Review of standards for energy performance of chiller systems serving commercial buildings,” *Energy Procedia*, årg. 61, s. 2778–2782, 2014.
- [12] “JRC Photovoltaic Geographical Information System (PVGIS).” Accessed: 2025-05-07, European Commission, Joint Research Centre (JRC). URL: https://re.jrc.ec.europa.eu/pvg_tools/en/#PVP.
- [13] “EN_Datasheet Tallmax_PE14H_2019_A_web.pdf.” Accessed: 2025-05-07, Trina Solar. URL: https://static.trinasolar.com/sites/default/files/EN_Datasheet%20Tallmax_PE14H_2019_A_web.pdf.
- [14] “SMA Sunny Highpower Peak1 75kW Inverter.” Accessed: 2025-05-07, DAT Solar. URL: <https://datsolar.com/en/p/inverter-sma-sunny-highpower-peak1-75kw/>.
- [15] C. Burgess. “Large-scale combined heat and power.” SERIES 13, MODULE 02: INDUSTRIAL CHP. In *Energy in Buildings & Industry*. Accessed: 2025-05-07, Energy Institute. URL: <https://www.energyinst.org/?a=538862>.

- [16] Queensland Competition Authority. “Final Determination: Regulated Retail Electricity Prices in Regional Queensland 2023–24.” Accessed: 2025-05-21.
- [17] Global LCA Data Access. “Thermal Energy and Electricity, Sugar Cane Bagasse, Open Input, Bagasse Combustion.” Life Cycle Inventory dataset. No publication date. Accessed: 2025-05-21.
- [18] I. M. O’Hara. “The Potential for Ethanol Production from Sugarcane in Australia.”
- [19] “Placa solar policristalina Trina Solar 340 W 144 celdas Split Max TSM-PE14H.” Accessed: 2025-05-07, Dismad Shop. URL: <https://dismadshop.com/es/policristalinas/151-placa-solar-policristalina-trina-solar-340-w-144-celulas-split-max-tsm-pe14h.html>.
- [20] “SMA Sunny Highpower Peak1 75-10.” Accessed: 2025-05-07, Tecno Logistica. URL: <https://tecno-logistica.it/negozio/it/trifase-10-100-kw/248-sma-sunny-high-power-peak1-75-10.html>.
- [21] D. Generazzo. “6.8 Rules of Thumb for CHP Engineering and Installation Costs.” From the CHP Applications Guide. Accessed: 2025-05-07. URL: <https://understandingchp.com/chp-applications-guide/6-8-rules-of-thumb-for-chp-engineering-and-installation-costs/>.
- [22] “Cost of Current Cooling Technologies.” Accessed: 2025-05-07, Enersion. URL: <https://enersion.com/tips-tricks/cost-of-current-cooling-technologies/>.
- [23] “Electricity Rates for Duquesne Light.” Utility pricing information. Accessed: 2025-05-07, Duquesne Light. URL: <https://c03.apogee.net/mvc/home/hes/land/el?utilityname=duquesneelight&spc=cel&id=19018>.
- [24] “Lithium Battery Freedom Lite 80kWh.” Accessed: 2025-05-07, Suministros del Sol. URL: <https://suministrodelsol.com/en/freedom-lite-lithium-battery/414-lithium-battery-freedom-lite-80kwh.html>.
- [25] “ZCS AZZURRO - INVERTER DI STRINGA TRIFASE 3PH 100KTL-V4/10KTL-V4.” Technical datasheet. Accessed: 2025-05-07, ZCS Azzurro / Tecno Logistica. URL: https://tecno-logistica.it/negozio/it/index.php?controller=attachment&id__attachment=614.
- [26] “Cheaper Home Batteries Program.” Last updated: 20 May 2025. Accessed: 2025-05-21, DCCEEW. URL: <https://www.dcceew.gov.au/energy/programs/cheaper-home-batteries>.
- [27] Queensland Competition Authority. “Final Determination: Regulated Retail Electricity Prices in Regional Queensland 2023–24.” Accessed: 2025-05-21, Queensland Competition Authority. URL: <https://www.qca.org.au/electricity/retail/prices/2023-24/determination-and-documents/>.
- [28] Australian Energy Market Operator (AEMO). “Electricity Pricing in the National Electricity Market.” Accessed: 2025-05-21. URL: <https://aemo.com.au/en/energy-systems/electricity/national-electricity-market-nem/market-operations/pricing>.
- [29] “Case Study No.42: 105 t/h Bagasse- / Coal-fired Boiler for Cogeneration.” Accessed: 2025-05-21, John Thompson (ACTOM Pty Ltd). URL: <https://www.johnthompson.co.za/wp-content/uploads/2020/09/case-study-42-105-th-bagasse-coal-fired-boiler-for-cogeneration-jul19.pdf>.